“Report: Census Analysis and Recommendations for Land Development and Investment in Town Infrastructure”

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➢ Overview

Census is a survey conducted in the UK every ten years to obtain information of the population, in this case we have been given a fictional census of a town between two major cities. The local government must decide the utilization of an open plot of land which is vacant and the other major investments should be planned because of population demographic and economic requirement(Ilyas and Rekatsinas, 2022). The information offered covers house numbers, street names, occupants' details, other information about marital status, occupation, diseases and even religion.

This report provides analysis to determine:

1. As for the plot of land it is still vacant, what should be constructed on the vacant piece of land.
2. To what extent are infrastructure and social services that are more important for investment?

➢ Data Cleaning

Here's a refined breakdown of my data cleaning process and suggestions for improvement**:**

	Unnamed: 0	House Number	Street	First Name	Surname	Age	Relationship to Head of House	Marital Status	Gender	Occupation	Infirmity	Religion
0	0	1.0	Wembley Ways	Katie	Whitehouse	62	Head	Single	Female	Medical technical officer	None	Christian
1	1	NaN	Wembley Ways	Colin	Singh	64	Partner	Single	Male	Colour technologist	None	Christian
2	2	2.0	Wembley Ways	Marie	Brennan-Welch	29	Head	Single	Female	Community arts worker	None	None
3	3	2.0	Wembley Ways	Tom	Jones	31	Partner	Single	Male	Recruitment consultant	None	None
4	4	2.0	Wembley Ways	Jenna	Brennan-Welch	12	Daughter	NaN	Female	Student	None	NaN
5	5	3.0	Wembley Ways	Kirsty	Powell	56	Head	Single	Female	Naval architect	None	None
6	6	3.0	Wembley Ways	Edward	Powell	25	Son	Single	Male	Retail manager	None	None
7	7	3.0	Wembley Ways	Jeffrey	Powell	21	Son	Divorced	Male	University Student	None	None
8	8	4.0	Wembley Ways	Scott	Sutton-Robinson	52	Head	Single	Male	Unemployed	None	Christian
9	9	NaN	Wembley Ways	Leah	Sutton-Robinson	23	Daughter	Divorced	F	Travel agency manager	None	Christian

Figure 1: Data preview before Cleaning

We used the setting to make them not interpreted as NaN in the frame as we need 'None' values especially in the religion column to be accepted as such but as some entries that are empty. Further, we also ignored the rows wherein the first name and surname were blank as they were nearly bare records or entries that had insufficient information that could be tied to the identity of the person in question(Chauhan and Samuel, 2024).

1. Handling Missing Values

- **First Name and Surname**: We pruning out records where both First Name and Surname were blank which was a wise decision because such records are not of much help.
- **House Number**: As for the house number, we simply copied them with reference to the Surname of the surrounding rows – a very useful strategy that helps to have the logical flow within the householders.

- **Street Name**: For Street Name, we utilized forward fill (ffill) because the house numbers in a street are normally similar and studied the sequential dependence among them.
- **First Name of Occupant**: The cases in which the first name was missing were handled by replacing them and with an empty string which is allowed especially where last name and other parts of the address give enough information about the occupants of the house.
- **Surname**:Regarding missing Surname values we used the mode based on House Number and Street in order to keep the family integrity.

2. Age of Occupant

- We thus used the mode of the age to fill in for missing values in age.

3. Relationship to Head of Household

- The rationale behind completing the field called Relationship to Head of Household was based upon the age and composition of the household. We ensured:
    - We only substituted values for those over 18 years of age only , as Head of Household where no other Head exists in the household.

4. Marital Status

- **Minors**: We automatically set one who is below 18 years as the status of Minor.
- **Married**: Regarding the missing data, we kept marital status as 'Married' for persons with relationship as Husband or Wife which we deemed allowable.

5. Gender

- Our gender-filling strategy was based on the Relationship to Head of Household field which matched the dataset correctly.
- As mentioned before, this mapping is used to identify the gender based on the Family relationships presented in the given data set. For example, "Husband" is defined as "Male," and "Wife" as "Female." "Son" and "Grandson" are both "Male," while "Daughter" and "Granddaughter" are both "Female." Using this method, some missing gender values are easily filled in by using gender information derived out of the relationship information given in the data-set(Soren, 2023).

6. Occupation

- Thus, we omit the rows which contain the empty occupant.

7. Infirmity

- We removed rows that contained the variable occupation but it is with blank values.

8. Religion

- For missing religion we imputed the mode of Religion for each household based on House Number and Street to maintain household consistency. This was reasonable provided there was no other evidence to the contrary. Also, one important thing we did is we replaced 'Jedi' religion as 'None' as Jedi is not a real religion but a fake religion used as a joke or protest.

9. Grouping Occupations

- Concerning the Occupation field, we categorized it into general headings like Retired, Universtiy student, student, unemployed and Child, which made it easier to code. We made the grouping appropriate to capture the complexity of the occupational landscape as much as possible, with finer divisions if required(Lipovetsky, 2020).

	House Number	Street	First Name	Surname	Age	Relationship to Head of House	Marital Status	Gender	Occupation	Infirmity	Religion
0	1.0	Wembley Ways	Katie	Whitehouse	62	Head	Single	Female	Medical technical officer	None	Christian
2	2.0	Wembley Ways	Marie	Brennan-Welch	29	Head	Single	Female	Community arts worker	None	None
3	2.0	Wembley Ways	Tom	Jones	31	Partner	Single	Male	Recruitment consultant	None	None
4	2.0	Wembley Ways	Jenna	Brennan-Welch	12	Daughter	Minor	Female	Student	None	None
5	3.0	Wembley Ways	Kirsty	Powell	56	Head	Single	Female	Naval architect	None	None
6	3.0	Wembley Ways	Edward	Powell	25	Son	Single	Male	Retail manager	None	None
7	3.0	Wembley Ways	Jeffrey	Powell	21	Son	Divorced	Male	University Student	None	None
8	4.0	Wembley Ways	Scott	Sutton-Robinson	52	Head	Single	Male	Unemployed	None	Christian
9	4.0	Wembley Ways	Leah	Sutton-Robinson	23	Daughter	Divorced	Female	Travel agency manager	None	Christian
10	5.0	Wembley Ways	Dawn	Wilson	48	Head	Widowed	Female	Unemployed	None	Christian

Figure 2: Data preview after Cleaning

➢ Core Analysis for Census Data-Based Decision-Making

This analysis is intended to enable crucial decision making in regards to development of land and infrastructural investment of a town(Prema, Raju and Ramya, 2022).

1. Age Distribution and Population Growth

- **Population Growth**: The birth rate is 19.82 birth rates per thousand population while the death rate remains comparatively low at one rate per thousand population Only 1 death per 1000 population. This implies that there is a continuous base population growth in the natural course of population growth while a huge population influx of young people.

- **Age Distribution**: Approximately 40-45 years old is the average employment, this evidences that the subject workforce is still capable. It is also evident that the number of young persons below the age of 18 and students pursuing their studies at universities is also very high meaning that in the future the population of the nation is likely to increase since it is not only stable but also comprises of young people.

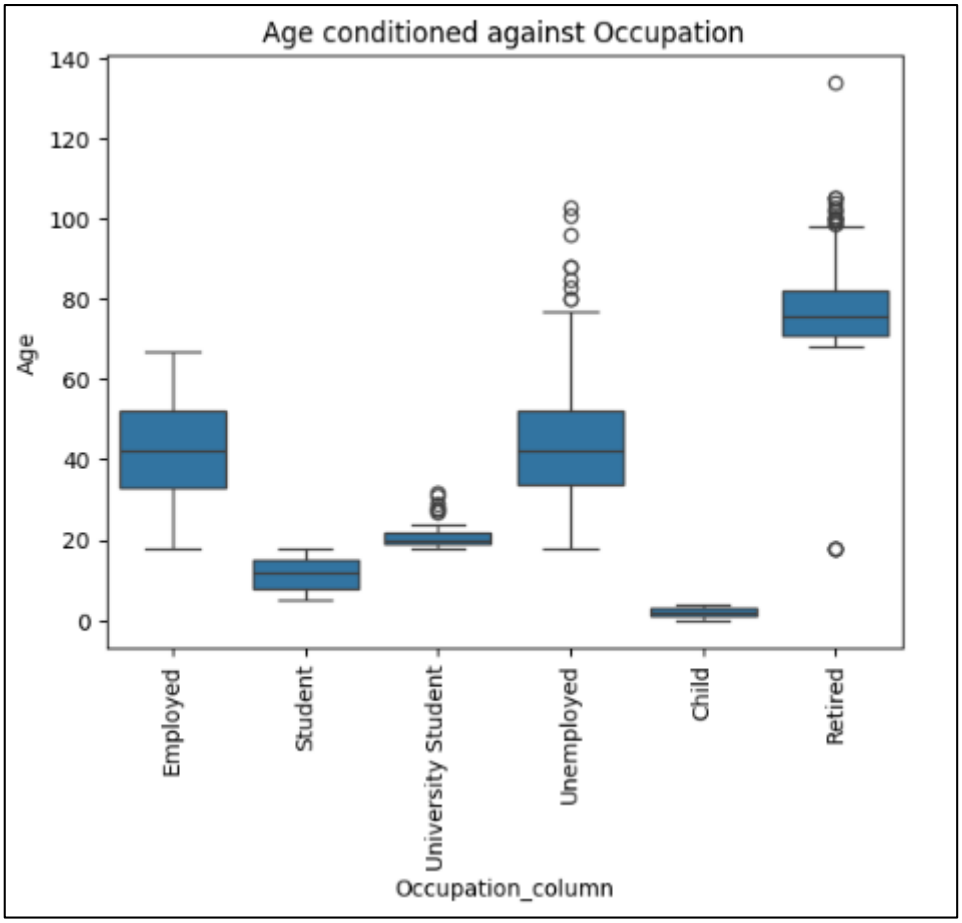

Figure 3: Age conditioned against occupation

- **Retirement Age Population**: A small proportion of people belong to the retirement group while the average age is 70 years among them. On the same note, the population belonging to the elderly group is expanding albeit at a more slower rate than the working-age and the youth.

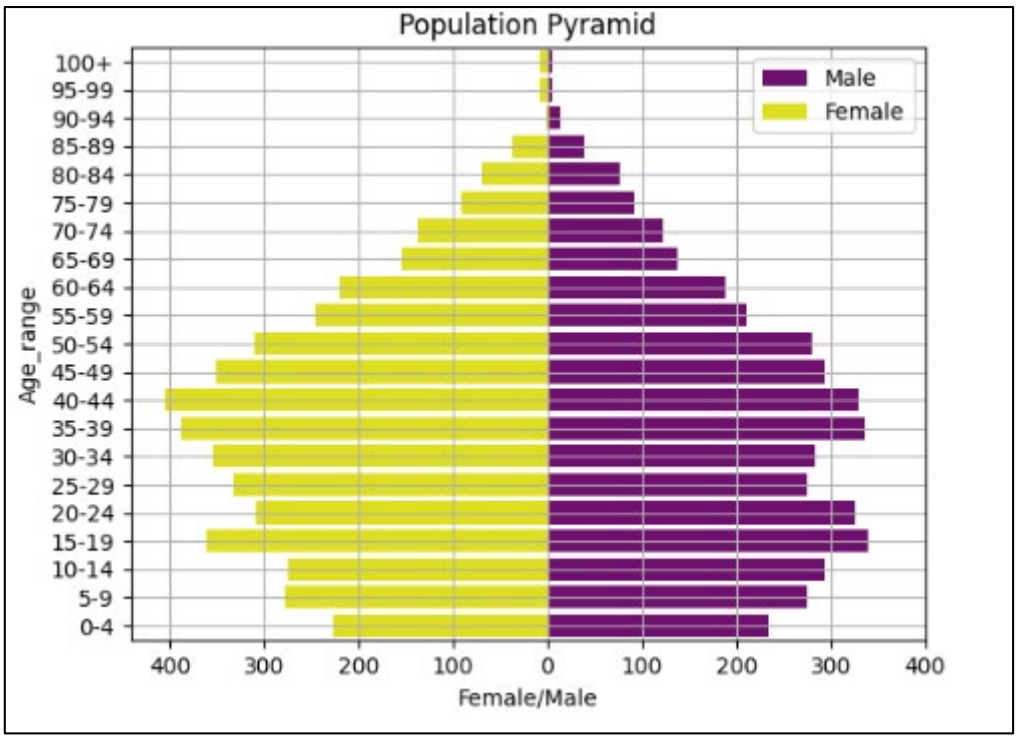

Figure 4: Age pyramid

Key Insight: The total population is increasing and the population pyramid suggests that there is population density in the young people as well as the elderly. This justifies choices that make provision for the current increase in housing demand and could serve to accommodate future elderly population if called for.

2. Unemployment Trends

- **Unemployment Rate**: Some of the features include; The unemployment rate is at 6.48%, which is moderate. The unemployed individuals are of different age, though the mean age is approximated to be 45 years. It is during middle adulthood that unemployment is high with fewer cases of unemployment among the young people.

- **Employment Categories**: Employment rate in the region is above 54% and university students contributing to nearly 6.12% of the population. This suggests that this town has a stable working force but at the same time a significant number of people may require retraining and skills upgrade.

Occupation_column	count
Employed	4727
Student	1632
Retired	810
Unemployed	566
University Student	534
Child	461

Figure 5: Occupation value counts

Key Insight: Support the employment and training activities might better meet the requirements of middle-aged unemployment persons and enhance their employment status and even effect the economy development.

3. Religious Affiliation

- **Dominant Groups**: The largest religious group in the town according to the census is the one with "none" which is more than 3500 people. Christians and Catholics are the two next follow religious groups, however, their populations are already aging which may means that those group can be stagnated or decrease.

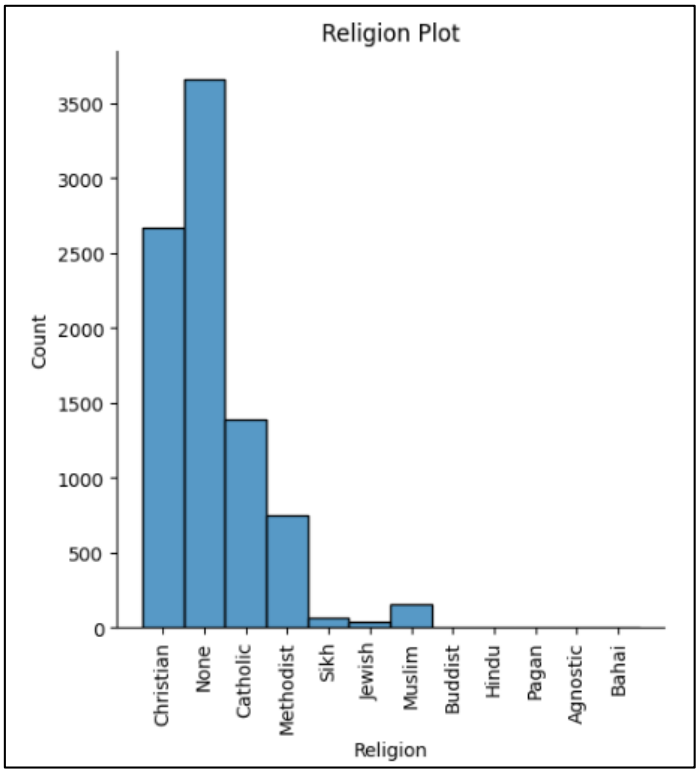

Figure 6: Graph for Religion Value counts

- **Growth in Minority Religions**: Younger representation there is found in Muslim, Sikh, and Jewish populations, which may suggest a possibility of the increase in these minorities, although, the numbers are small.

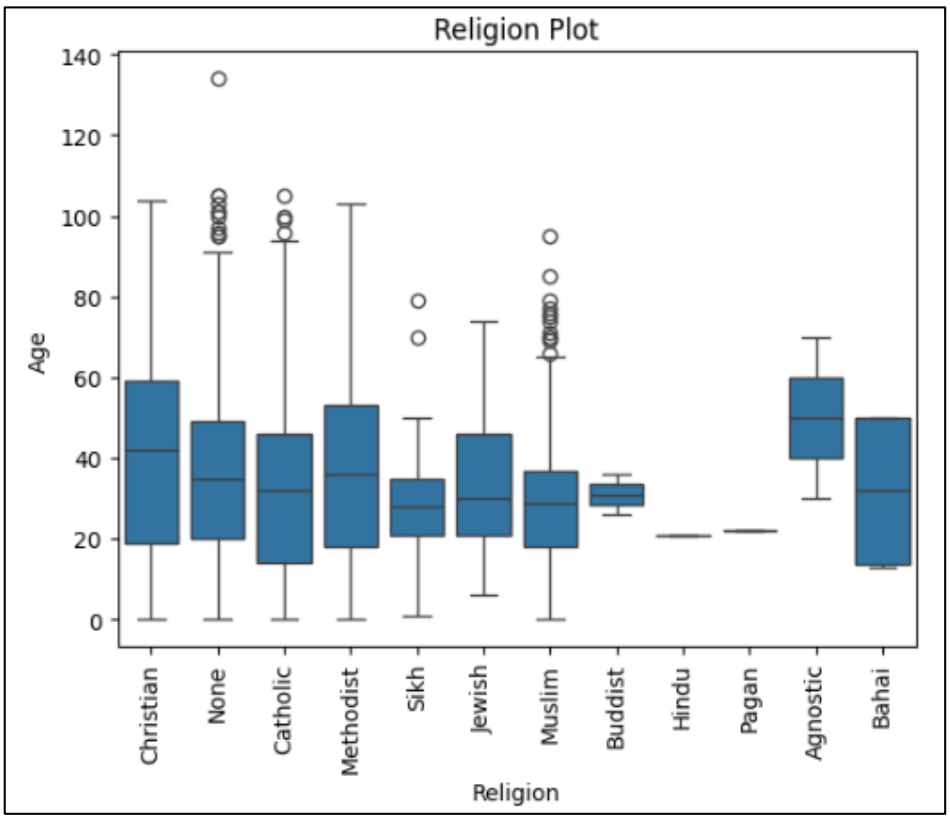

Figure 7: Box Plot for Age against Religion

Key Insight: The current data does not show the need for construction of new religious buildings however the monitoring of growth of the minority religions for instance the Islamic faith may consider new infrastructure.

4. Marriage and Divorce Rates

- **Marriage vs. Divorce**: The rates of marriages are low at 294.04 per 1,000 people and on the other hand the divorce rate has established 92.55 per 1,000 people. Consequently, for every divorce, there are about 3.18 marriages. There is a general distribution of marriage ages, whereas divorce is more likely to occur between the age 30 and 60.

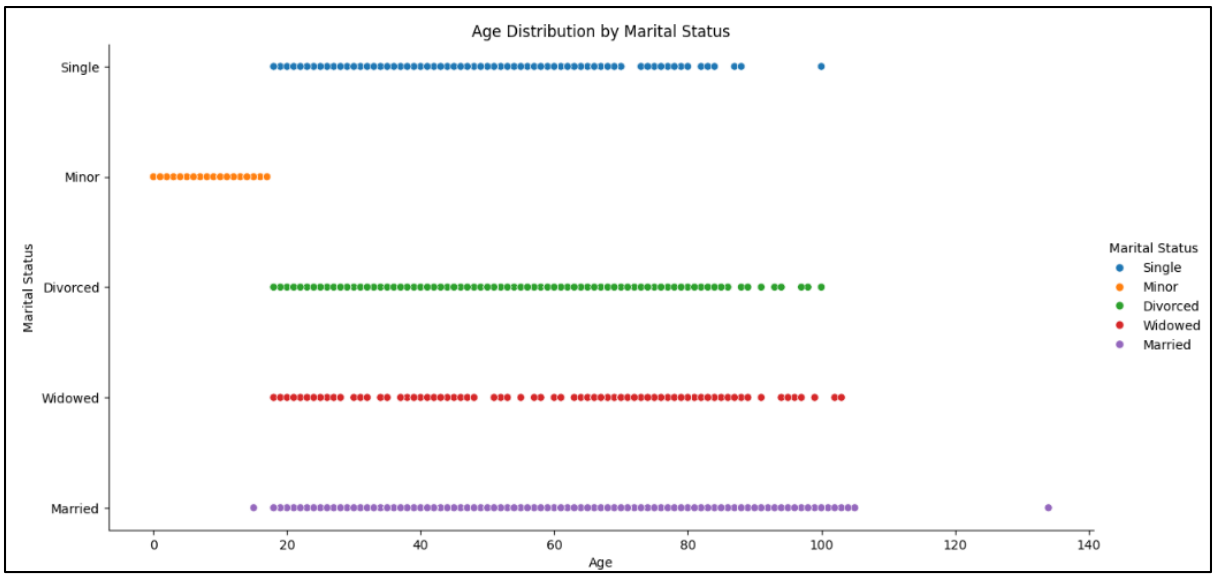

Figure 8: Age distribution by marital status

Key Insight: A high proportion in the marriage advancement means that housing development should accommodate families as those in their middle ages and who are married would prefer family homes, which are stable and secure.

5. Occupancy Levels

- **Household Occupancy**: On the typical family size in the households, it is estimated that a normal size is about 2.64 persons most of the populations are single or two headed household. The proportion of the larger households with four and more people is rather small – 10.71% of the population. Meanwhile, 27.22 per cent of the households are identified as under-occupied household**.**

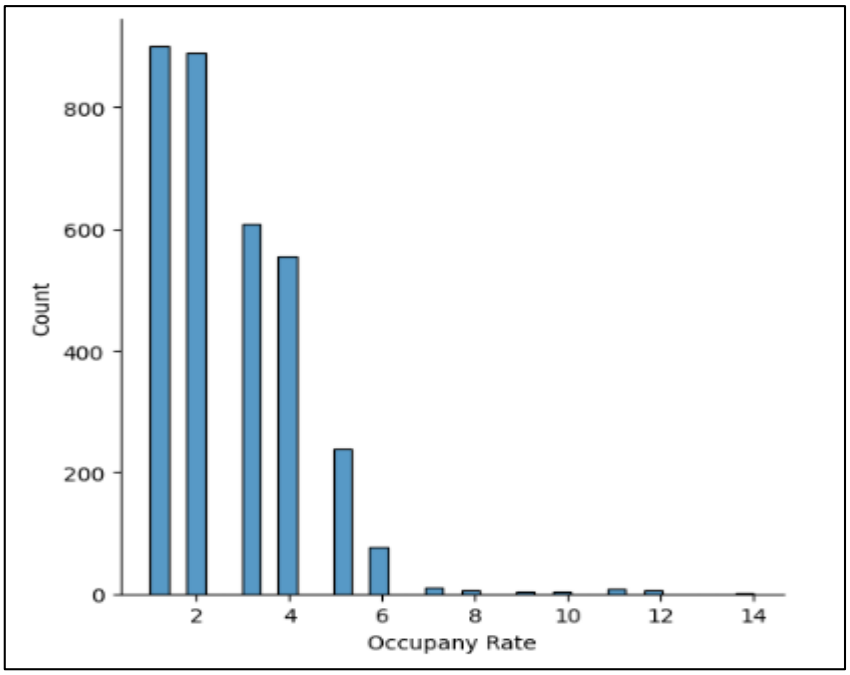

Figure 9: Graph for Occupancy Rate per house

Key Insight: The percentage of households in units with more rooms than needed is drastically low and indicates that populations density, would be better served through high density housing rather than low-density ones in the town.

6. Commuter Population

- **Commuting Trends**: About 60% of the population have to use the vehicles, mostly people with jobs, and students attending university in other cities. These high rates cause congestion of local roads and also exert pressure on other means of transport.
- In order to estimate how many of the commuters were there, we consider those with occupation as "Employed" and "University Student" Then add the two together and divide by the total population. Multiplied by 100 yields for the percentage of commuters and here it is 60 percent.

Key Insight: Thus, constructing a train station will help to reduce the traffic problem and cater for the number of commuters.

7. Birth and Death Rates

- **Birth Rate**: New families formation is relatively healthy in the town as evidenced by 19.82 new families per 1000 people.
- Birth rate is defined as a number of persons of half a year or less old divided by total population or 8730 and multiplied by 1 000. The result as estimated is around 19 82 births per one thousand population.
- **Death Rate**: The low death rate of 1.72 per 1,000 people further augments the growth of the population.
- To estimate deaths we compare consecutive age groups in order to see how many people are missing from the next age group as they may be dead. Treat these people as dead and come up with death rates as a percentage by using the number of deaths and the population number then divided it by 1000.

Key Insight: To justify expenditure on schooling and other general structures there must now be more families and school going children in the town than before.

➢ Task (a): A plot of land is empty, and the issue is what should be constructed on it?

Options:

Criteria Analysis:

1. High-Density Housing

High density housing is applicable most of the time especially when the population of a given region is booming. Key factors to consider include:

- **Population Growth Rate**: population growth rate calculated from the previous formula above is around 18.1%, indicating positive growth.
- **Age Distribution**: It has a large population of the youthful population in the 0-17 and the 18-34 age groups which implies that there will be a need for housing in future.

Age_range	Male	Female
0-4	234	-227
10-14	294	-275
100+	6	-9
15-19	339	-362
20-24	326	-309
25-29	276	-333
30-34	283	-354
35-39	337	-388
40-44	329	-405
45-49	293	-351
5-9	276	-279
50-54	280	-311
55-59	210	-246
60-64	188	-221
65-69	138	-155
70-74	122	-137
75-79	92	-91
80-84	77	-70
85-89	39	-38
90-94	13	-2
95-99	6	-9

Figure 10: Graph for Age range and no of individuals in each

- **Household Formation**: Statistics proved that most heads of households are within the age bracket of 30-59 years meaning that almost all are middle aged people.

Age_range	
40-44	367
35-39	352
45-49	328
50-54	298
30-34	288
55-59	255
25-29	223
60-64	216
65-69	197
70-74	187
20-24	171
75-79	134
80-84	104
85-89	60
15-19	59
95-99	12
100+	9
90-94	7

Figure 11: Age range for heads of houses

Based on the realization that there is a positive population growth rate and youthful families dwelling more in the city, high density housing can be one reasonable solution to the increase in family units.

2. Low-Density Housing

Low density housing should be justified in that the population is wealthy and demand more space in their homes. Key factors:

- **Income and Occupation**: 54.15 percent of the population is either employed and a small fraction, 6.48 percent is jobless. The category of 'employed' is prominent in the census data and yet the index does not suggest that the society is prosperous.

Occupation_column	proportion
Employed	0.541466
Student	0.186942
Retired	0.092784
Unemployed	0.064834
University Student	0.061168
Child	0.052806

dtype: float64

Figure 12: Occupation proportion of populaation

- **Marital Status**: 1,212 respondents are currently married stating family oriented living while 691 of them have children. But they fail to show that there is much of a call for more expansive residences.

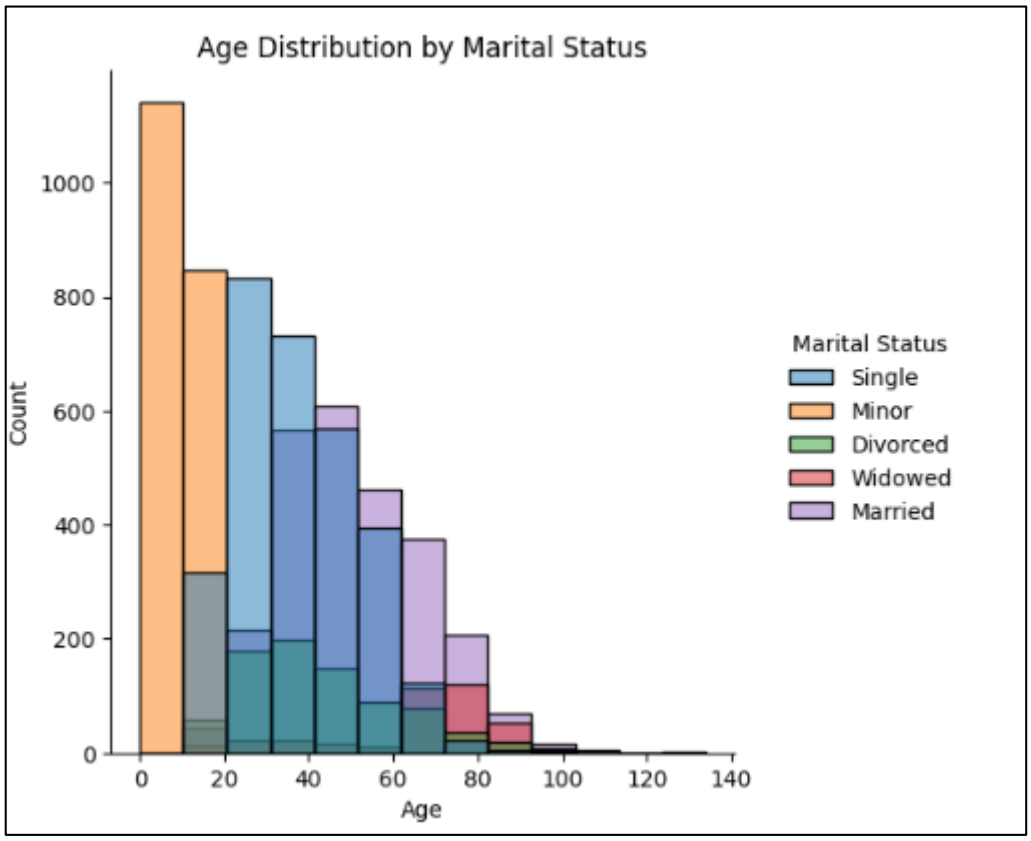

Figure 13: Age distribution by marital status

It might be false as there is little or no proof of prosperity; hence, low-density housing may not be a consideration.

3. Train Station

Some of the factors which may require the construction of a train station are many people in a particular area who may be in need of the transport facility. Key factors:

- **Commuter Data**: About 60% of the people are traffic commuters are employed ones or students going to the University.

A train station could ease the pressure off commuting and can be an effective option to resolve road nuisance, not to mention with such a high commuters' percentage.

4. Religious Building

Religious building could require a new construction depending on the religion and the number of religious believers. Key factors:

- **Religion Data**: Hence, the most represented religion is "None" which comprises of 3,664 persons whereas the Christians are 2,673. Muslims, Sikhs and Jewish are other religions stated to be growing but they are not many to warrant construction of new religious facilities.

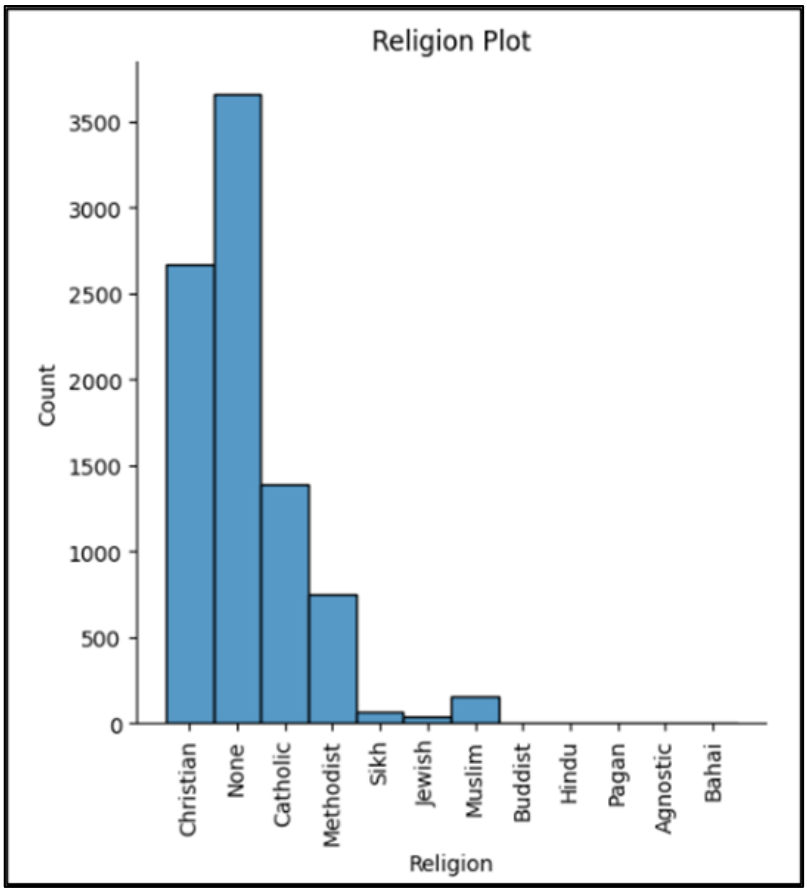

Figure 14: Religion Value count of population

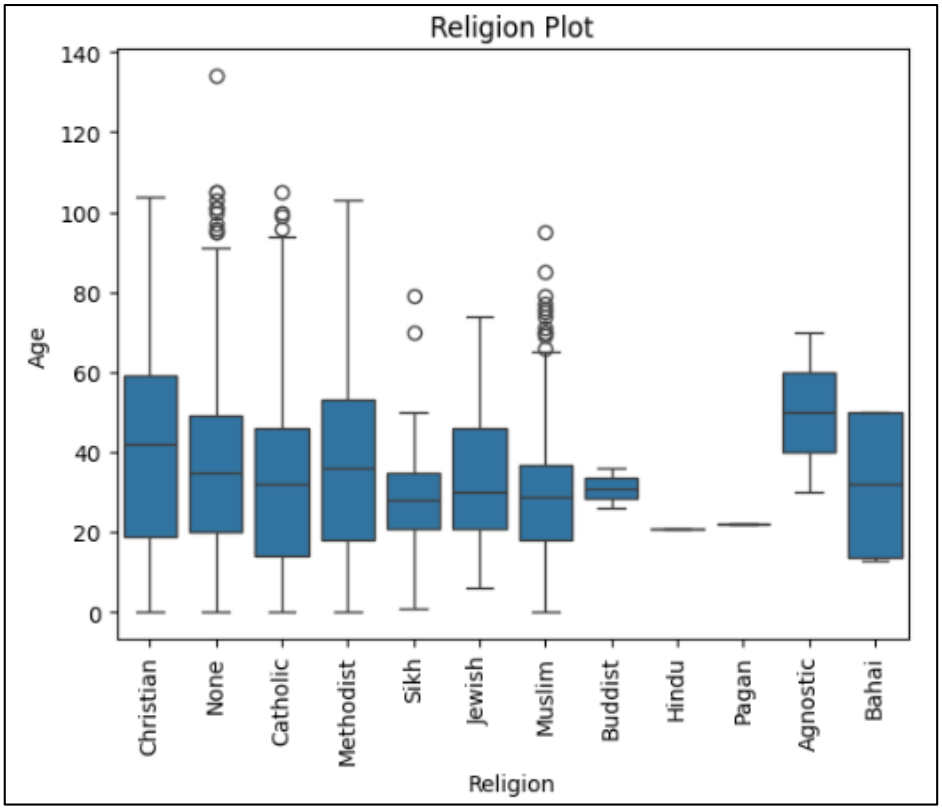

Figure 15: Box plot for age against religion

It can be noted that at the moment, there is no an overwhelming need for new religious structures.

5. Emergency Medical Building

From these, one might need a minor injuries center if the respective rates are high or in case of potential pregnancies. Key factors:

- **Infirmity Data**: The majority of people stated that they have no impairments (8,669); a limited number of respondents indicated that they have some form of disability, are blind or have other types of impairments.

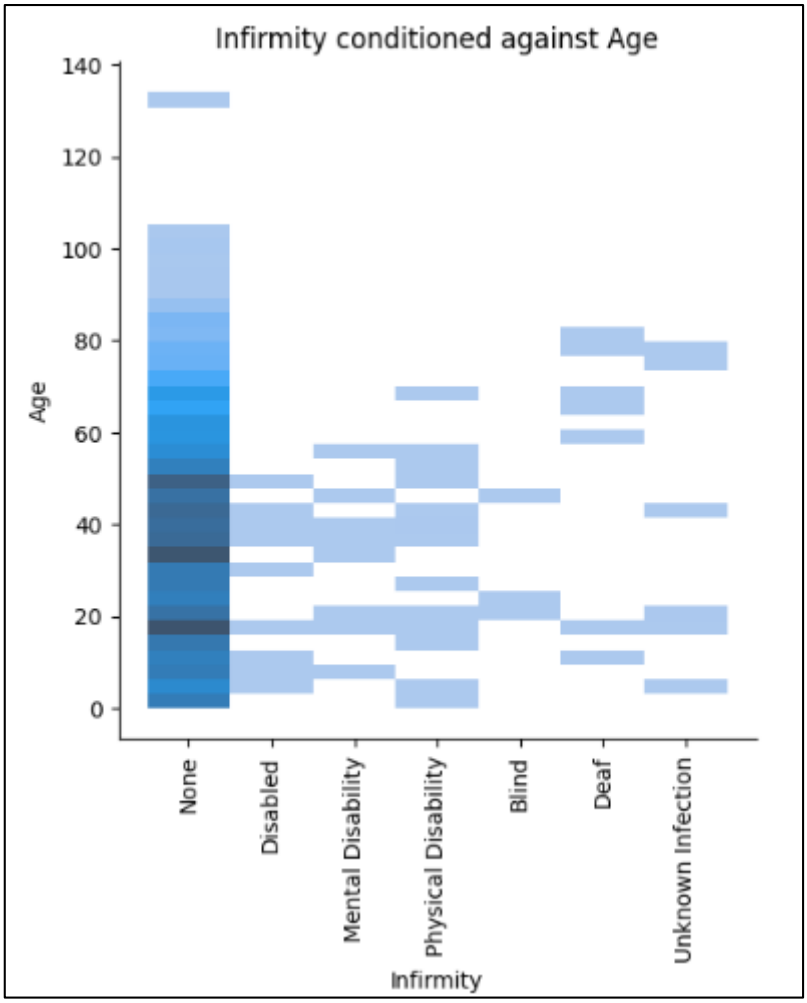

Figure 16: Infirmity condition against age

- **Birth Rate**: The numbers are as follows, the birth rate equals to 19.82 per 1,000 people, which indicates continuing migration by young families, which may need medical help in connection with pregnancies.

They added there is moderately strong evidence toward the necessity of improving the availability of hospitals further taking into account the birth rates.

Recommendation for Task (a): Construct a train station to cater for the 60% of the people who commute to work. There is also capacity consideration; high-density housing may also be considered with reference to the increasing population.

➢ Task (b): Which one of the options below should be funded?

Criteria Analysis:

1. Employment and Training

- **Unemployment Rate**: The unemployment rate is 6.48%," while the largest number of unemployed persons was aged between 30 and 60 years.

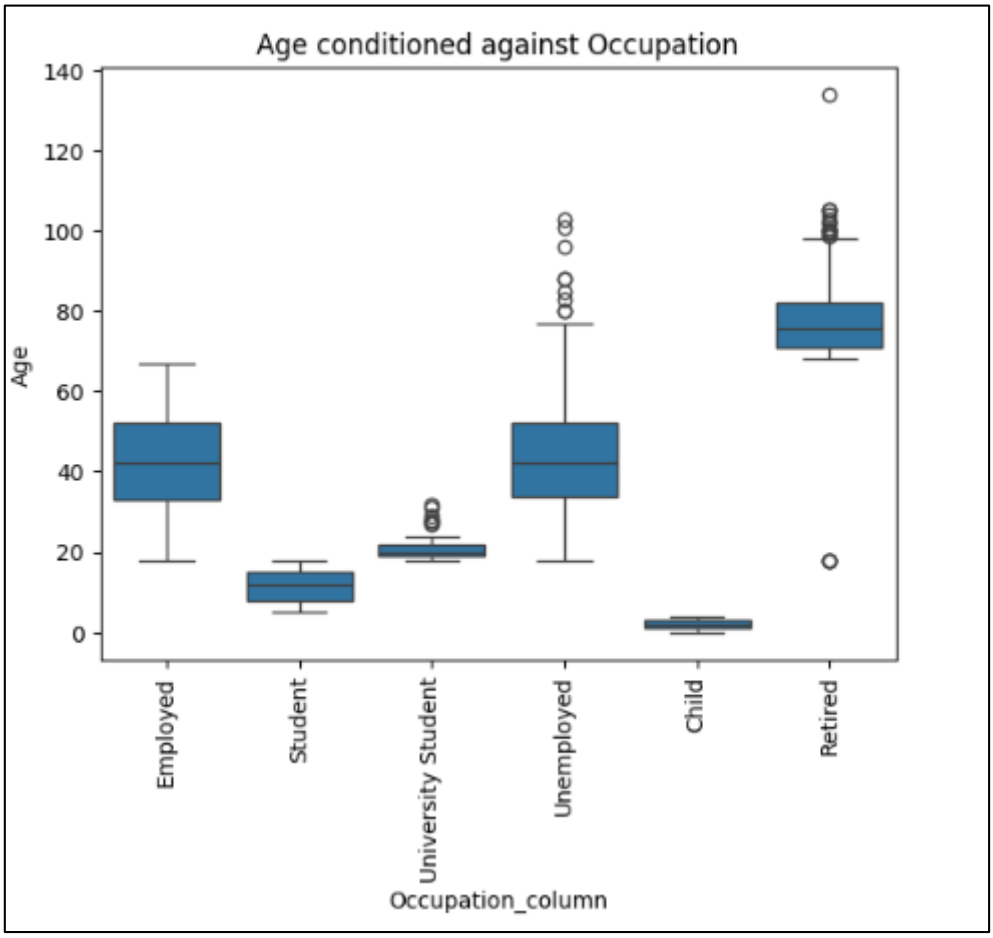

Figure 17: Age conditioned against occupation

- **Occupation Data**: The highest number of employment-seekers falls under this category and there are as many as 4,727 of them. Nonetheless the unemployed persons (566) may likely benefit from any training programs launched in the company.

Occupation_column	count
Employed	4727
Student	1632
Retired	810
Unemployed	566
University Student	534
Child	461

Figure 18: Occupation value count of population

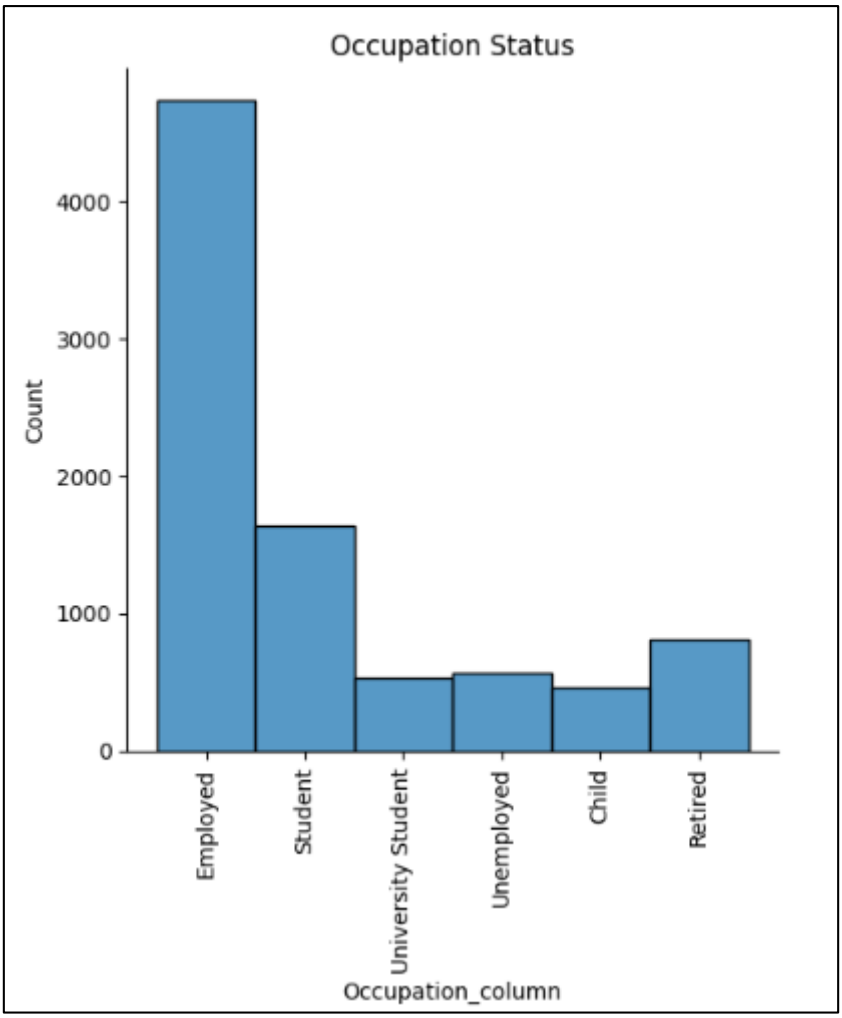

Figure 19: Graph for occupation of population

These places could include employment and training programs, where all the unemployed, particularly the youthful persons below the age of 50 could be trained to acquire new skills in the market.

2. Old Age Care

- **Elderly Population**: The aged people in the town are 945 of age above 65 and the dependency ratio is 0.16 (That is 16 elderly to 100 Working age people).

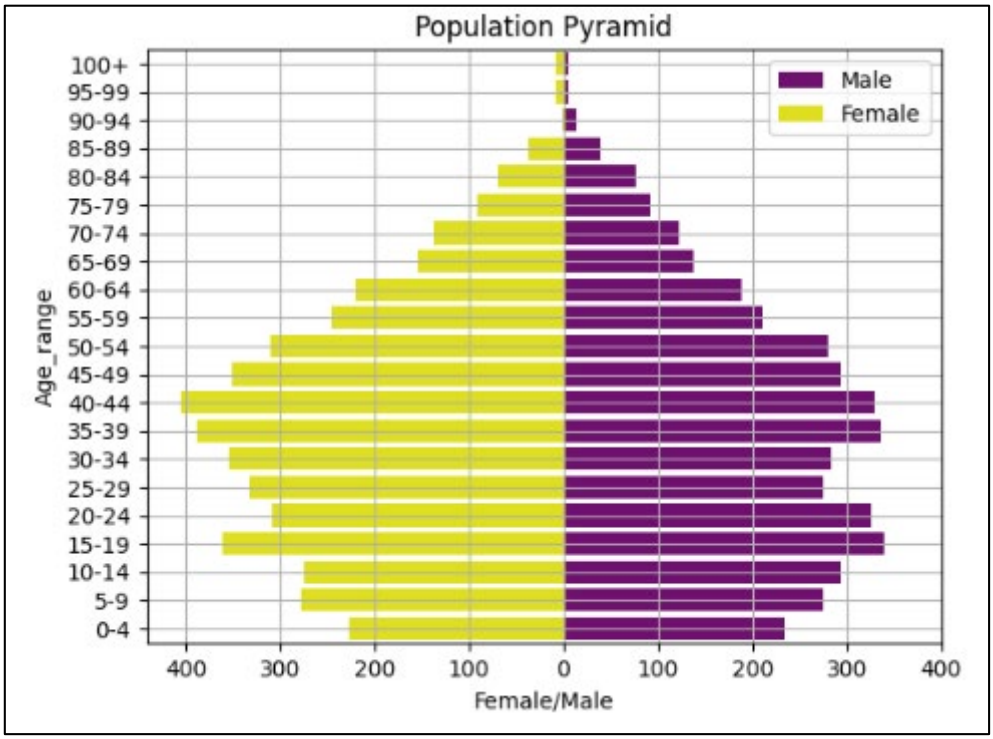

Figure 20: Age pyramid for age distribution

- **Infirmity Data**: Pethora's disabilities, similar to other infirmities, tend to be age-related and may include mental illness.

Infirmity	count
None	8669
Physical Disability	18
Disabled	12
Deaf	9
Unknown Infection	9
Mental Disability	8
Blind	5

Figure 21: Infirmity value counts of population

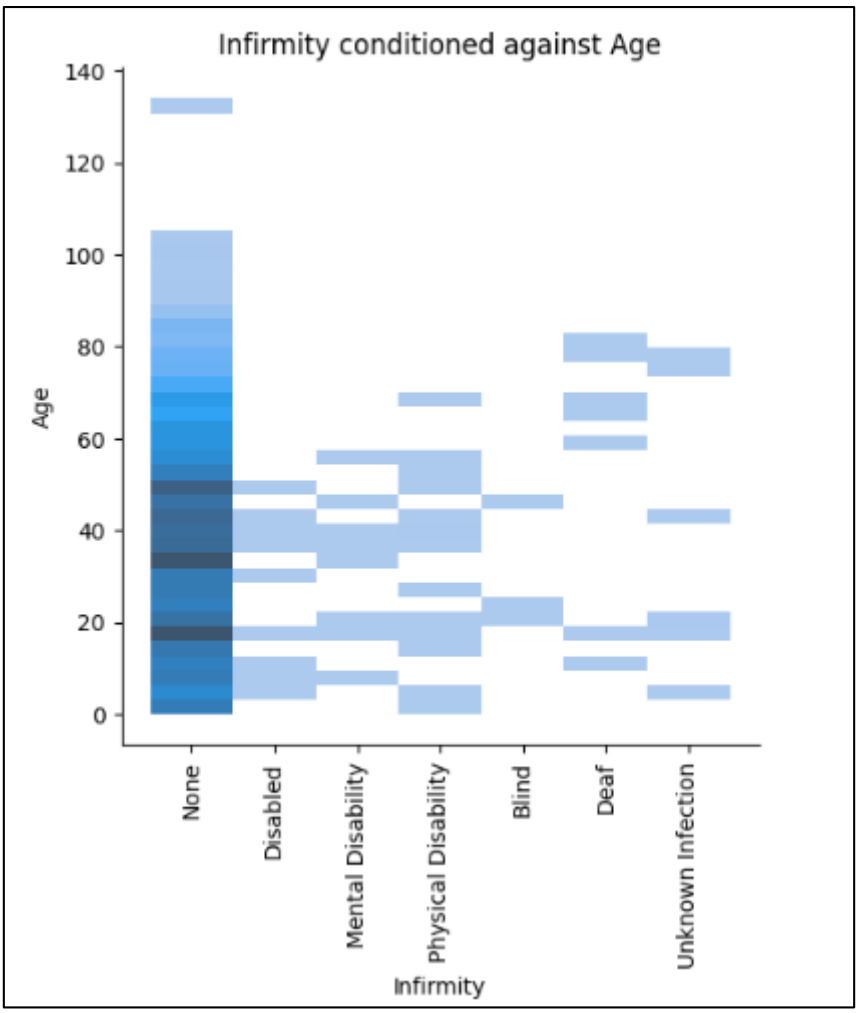

Figure 22: Infirmity condition against age

There will be need for old age care as the population ages and this is not necessarily be a primary consideration as compared to any of the other options.

3. Increase Spending for Schooling

- **Children Population**: The birth rate is moderate (19. 82/1,000) and the child population remains almost unchanged: 461 houses for families with children having none of them under the age of 4.
- **Married Couples with Children**: 549 parents have school going children proving that we have at least a stable and possibly even growing school going population.

Age_range	Male	Female
0-4	234	-227
10-14	294	-275
100+	6	-9
15-19	339	-362
20-24	326	-309
25-29	276	-333
30-34	283	-354
35-39	337	-388
40-44	329	-405
45-49	293	-351
5-9	276	-279
50-54	280	-311
55-59	210	-246
60-64	188	-221
65-69	138	-155
70-74	122	-137
75-79	92	-91
80-84	77	-70
85-89	39	-38
90-94	13	-2
95-99	6	-9

Figure 23: Age distribution for population

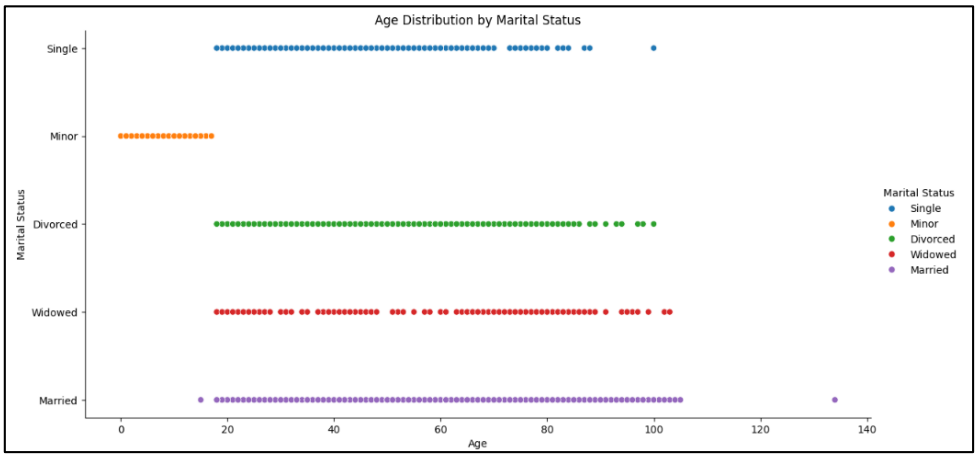

Figure 24: Age distribution by marital status

Extra spending for schooling was expected to be done with the more increasing population of children below the age of 18.

4. General Infrastructure

- **Population Growth**: The population growth rate is 18.1%, it is stated that general infrastructure will require to grow in connection with the increase in the size of the town.
- **Housing Data**: Current demographic shows that 27.22 percent of houses are under-occupied, but of course the upcoming population shows that there may be need for expansion and this brings along requirements such as infrastructural developments such as waste management, road maintenance among others.

```
Average household size per street:
                   Street  Average Household Size
0    Aethelred Manorhouse               12.000000
1             Albion Fort                5.000000
2              Allen Well                3.024390
3         Anderson Street                2.125000
4        Aquarius Rectory               12.000000
..                    ...                     ...
100          Wyatt Avenue                3.400000
101          Yates Summit                3.076923
102         Yellow Street                2.696429
103          Zigzag Drive                1.654676
104         Zigzag Street                3.275000

[105 rows x 2 columns]
```

Figure 25: Average household size per street

```
Total population: 8730
Total number of households: 3306
Average population per household: 2.64
```

<u>Recommendation for Task (b): Raise expenditure for schooling as the population of school going children rises and for future future expansion of schooling facilities as the town grows in size.</u>

➢ Conclusion:

In the case of the unoccupied plot of land, the first concerning issue should be the construction of a train station to cater for this highly commuting town. As for the expenditure, there is a need to increase spending for schooling due to the expansion of the size of school going age children in the population.

➢ References

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